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Explainable Reinforcement Learning for Power Grid Operations

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Submitted in partial fulfillment of the requirements for the MSc degree in Advanced Computing of Imperial College London

June 2026

Contents

Chapter 1

Introduction

1.1 Motivation

The second industrial revolution brought about the rise of electrification for powering critical infrastructure and satisfying the needs of the general public. Energy has therefore served as a vital component for meeting the demands of continual social and technological developments. Alongside this, there has been an increased global emphasis on combating climate change [1], especially by the United Nations, who have incorporated it into its Sustainable Development Goals, namely SDG 13, which states that we must ”take urgent action to combat climate change and its impacts” in all actions and policies [2]. Thus, it has become necessary to rapidly evolve modern power systems to integrate and handle variable renewable energy (VRE) sources within the network for progress towards decarbonisation [3]. This large-scale transition poses significant operational challenges, requiring improved reliability and dynamic flexibility. This additionally places greater pressure on grid operators who must ensure stable operation. Undoubtedly, the intermittency of renewables such as wind power and photovoltaic solar power further significantly increases the complexity of transmission networks [4], with challenges related to quality, flow, stability, and balance impacting performance of these systems [5].

For tackling this, real-time grid topology optimisation has been noted as a cost-effective and dynamic method for handling the reconfiguration of grids to handle uncertainties and changes within the system [6]. Given the combinatorial nature and high-dimensional search space involved for this solution, Reinforcement Learning (RL) represents a natural solution. It has shown to be highly effective in tackling sequential decision-making problems in a plethora of domains, including but not limited to video games, robotics, and autonomous driving [7]. However, integration into real-world domains continues to remain limited by unresolved challenges within the RL domain related to compliance with physical and operational constraints, managing uncertainty, and combating partial observability [8]. Power grid operations further exacerbate these problems.

Recent research work has brought forward a reinforcement learning benchmark environment [9] to accurately simulate realistic power grid operations, and work has been completed to successfully evaluate reinforcement learning algorithms for

achieving strong survival performance. Despite certain reinforcement learning methods encompassing the ability to achieve high performance by learning an effective policy, the black-box nature of reinforcement learning policies raises significant safety concerns [10]. The lack of interpretability and explainability in its decision-making process hinders its seamless integration for deployment in the real-world alongside expert system operators. If its decision-making process cannot be understood or justified, grid operators would be reluctant to deploy these systems. As the complexity of the model increases, the difficulty of identifying potential failures and debugging issues while simultaneously monitoring and maintaining long-horizon goals increases in turn. Thus, transparency and trustworthiness are essential requirements for such safety-critical infrastructure.
Explainable Reinforcement Learning (XRL) is a subset of explainable artificial intelligence (XAI) that aims to improve upon this problem, ensuring explainability and interpretability, while preserving task performance. Therefore, there is a need to focus on reinforcement learning for real-time grid control, using XAI methods to ensure decisions that are interpretable and actionable for grid operators.

1.2 This Project

This project investigates whether interpretable reinforcement learning methods can produce controllers that are not only transparent, but also operationally reliable in power grid environments, whilst maintaining high operational metrics. The overarching research question that motivates this project is as follows:

> Do current state-of-the-art methods for training interpretable reinforcement learning policies preserve operational performance in power grid topology environments, and can decision tree-based approaches for maintaining intrinsic interpretability improve upon faithfulness, transferability, and usefulness without substantially reducing survival performance?

Our work aims to make the following contributions:

1. Mapping the landscape of what is possible within explainable reinforcement learning for power grid operations. This includes studying and implementing existing approaches such as decision tree extraction, policy distillation, and symbolic reinforcement learning methods that aim to make black-box models intrinsically interpretable, and evaluating their scalability and robustness.

2. Implementing and evaluating select interpretable reinforcement learning methods within the RL2Grid environment, focusing primarily on decision tree acquisition, comparing performance and interpretability against one another within topology-based grids.

3. Analyse the scalability and robustness of these approaches, particularly focusing on how these change with increasing size and complexity of environments, to identify the best methods.

By the end of the project, we aim to provide a novel framework that aims to improve the scalability of interpretable grid controllers through tree simplification, such that survival performance and human readability are both preserved.

Chapter 2

Background

2.1 Reinforcement Learning

Reinforcement learning is a subset of machine learning where an agent learns the best possible way of achieving its goal within an environment. [11]. To mathematically define this within the context of power grids, we will take the same definition as the environment established by RL2Grid [9], which serves as the foundation for the evaluation of this project's final contributions.
This environment is represented as a Markov Decision Process (MDP), which is defined as a tuple $(\mathcal{S}, \mathcal{A}, P, R, \rho, \gamma)$. Here, $\mathcal{S}$ and $\mathcal{A}$ are the state and action space sets respectively, P is the state transition probability distribution, ρ is the initial uniform state distribution, R is a reward function, and γ is the discount factor [9]. Due to the stochastic nature of MDPs, they obey the Markov property that the actions taken are completely dependent on the current state, with the past and future states being conditionally independent.
In policy optimisation methods, the agent learns a parameterised policy $\pi_\theta(a_t \mid s_t)$, which is the probability of selecting action a_t given state s_t at a certain time step t. In value-based methods, the agent instead estimates value functions such as $V_\pi(s)$ or $Q_\pi(s,a)$, which represent the expected discounted return obtained by following policy π from a given state or state-action pair. The objective is to learn a policy that maximises the expected discounted return:

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right]. \tag{2.1}$$

2.1.1 Policy Gradient Methods

Policy gradient methods [7] directly optimise a parameterised policy by estimating the gradient of the expected return with respect to the policy parameters. Rather than learning only a value function, these methods update the policy π_θ in the direction that increases the likelihood of actions that lead to higher estimated advantage. Policy improvement is done over a batch of sampled trajectories. [7]

Policy Gradient Methods allow for the optimal policy to be found by prioritising actions that maximise the reward through parameter tuning. There exist various methods of computing this, one of which is Proximal Policy Optimisation (PPO). Due to the availability of PPO within the RL2Grid benchmark, the next sub-section will delve into this.

Proximal Policy Optimisation

Proximal Policy Optimisation (PPO) is an on-policy actor-critic reinforcement learning algorithm that improves the stability of acquiring policy updates by optimising the parameters using stochastic gradient ascent [12]. Trajectories are collected using the current policy π_θ, which is then compared against the old policy $\pi_{\theta_{\text{old}}}$. $\hat{A}_t$ is the estimated advantage function at a certain timestep t, which is used to measure the improvement in the actions. To ensure that policy updates are minimal and therefore stable, the probability ratio $r_t(\theta)$ is clipped in the objective.

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \text{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t \right) \right]. \tag{2.2}$$

Here, $r_t(\theta)$ is the probability ratio, which is shown below, and ϵ is a tunable hyper-parameter. The value function parameters are simultaneously updated by aiming to minimise the loss function of the value function.

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}, \tag{2.3}$$

2.1.2 Value Based Methods

While policy gradient methods focus directly on the policy itself, value based methods derive the expected rewards from the state-action pairs that are chosen, in order to understand the best possible action to take given a state [13]. A value function is first estimated, and then repeatedly updated in each iteration to converge to the optimal value function that reflects the rewards captured in observations. Some examples of value based gradient algorithms are Q-Learning, SARSA, and Deep Q-Network (DQN). Once again, due to the presence of DQN within the RL2Grid benchmark, this algorithm will be explained in further detail in the next sub-section.

Deep Q-Networks

Deep Q-Networks (DQN) is a subset of deep reinforcement learning, where classical Q-learning is extended using neural networks in order to approximate the action-value function [14, 15]. The Bellman equation serves as the foundation of the target Q-value, which is incorporated into the loss function that is trained via deep learning techniques. The squared temporal-difference is minimised via gradient descent, and the formula [7] is as follows:

$$L(\theta) = \mathbb{E}\left[(y_t - Q(s_t, a_t; \theta))^2 \right], \tag{2.4}$$

$$y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-), \tag{2.5}$$

This algorithm is extremely useful for states where the actions are discrete, not continuous. By prioritising an ϵ-greedy policy, the balance between exploration and exploitation is handled, where there is a probability ϵ for the agent to select a random action, and probability $1 - \epsilon$ to select the action with the highest predicted Q-value.

DQN works due to the innovations of experience replay, target networks, and reward clipping. By breaking temporal correlations, ensuring that training does not occur on sequential states, the same experience can provide multiple levels of learning. Minibatch sampling is done during training to take advantage of this benefit, which is why past transitions are stored in a replay buffer.

Additionally, a separate network is created for computing the targets. This is because the target Q-value changes every update, making optimisation unstable. By using a separate target network $Q(\cdot; \theta^-)$ to store a network parameters, and copy the main network to the target network every couple steps (a constant), we can maintain this stability. Furthermore, the rewards are also clipped to ensure that stability is maintained. This is essential in domains such as power grids, where excessively large changes can have a drastic impact on the network.

2.2 Explainable Reinforcement Learning

Intervention is difficult when the safety, reliability, and explainability of an agent's decisions are unclear, especially when methodologies like neural networks are implemented alongside reinforcement learning frameworks for greater generalisation and overall improvement in performance. To tackle this, explainable reinforcement learning (XRL) allows users to gain an insight into decisions made and actions taken, improving transparency in ensuring that a model's behaviour is not only verifiable and acceptable by experts, but also explainable as to why a particular actions was selected in a given state.

As per the detailed survey conducted within this domain [16], existing work in this field can be classified under three categories, namely Feature Importance (FI), Learning Process and MDP (LPM), and Policy-Level (PL). As defined by this survey, Feature Importance takes into consideration the immediate context of the state that allows it to understand the action taken, whereas Policy-Level considers the a longer horizon of the model. Learning Process and MDP on the other hand learns from past experiences in order to gain an understanding of the current state of the Markov Decision Process.

Explainable reinforcement learning can be tackled using two broad categories; intrinsic interpretability, and post-hoc explanations. Intrinsic interpretability aims to design agents that are interpretable by construction, and include all the information regards to its decision-making process prior to choosing the optimal action. On the other hand, post-hoc explanation methods attempt to explain and justify the behaviour of already-trained black-box agents.

2.2.1 Policy Distillation

Although approaches like DQN allow for high-performing agents, these require intensive training, a large number of hyperparameters, and extensive network sizes during computation, which causes a large overhead on resources. Additionally, due to the nature of neural networks, which act as black boxes, users are unable to understand the decision-making process behind the actions suggested by the agents, making interpretation of not only the results, but also understanding its impact on the long-term goals of the model difficult to identify.

Policy distillation [17] addresses these two issues by establishing a new untrained network known as a student model S. The original model, referred to as the teacher model T, acts as the oracle policy in this architecture, whereby the student model is trained using state-action pairs generated by the teacher model. By imitating and approximating the actions of the expert policy, the student model learns to predict the action that the teacher model would have selected. This transfer of policies improves efficiency and speed with regards to decision-making as well as training.

The teacher model can be established using either policy gradient methods or value gradient methods, such as PPO and DQN respectively. This is particularly helpful when aiming to transfer policies, as additional black-box models are not required to be trained. Rather, we are able to inspect the policy directly and understand which features of the environment influence the actions of the agent, improving interpretability, and gaining a deeper understanding into being able to explain the optimal actions in the environment. Policy distillation provides an avenue into ensuring that model performance is optimised, while gaining the benefits of interpretability in the decision-making process. The student model can be of various types, including symbolic methods, rule-based models, or decision trees. Examples of approaches that are able to extract decision tree policies include Dataset Aggregation (DAgger) and Verifiable Reinforcement Learning via Policy Extraction (VIPER).

DAgger

Dataset Aggregation (DAgger) [18] is an imitation learning algorithm that improves upon simple behavioural cloning techniques by tackling the flaw of distribution shift, where the learned policy visits different states than the expert, which causes error compounding due to a lack of training data. By allowing the student policy to iteratively continue interacting with the environment while obtaining the expert labels from the teacher model, the student is able to retrain on an aggregated dataset that allows it to learn from trajectories that it previously did not have training data on. This improves its ability to be able to handle states that it previously would not have encountered if it were to purely be run on the original training dataset of the expert oracle. This approach is extremely useful in environments with large state spaces and actions possible, such as power grids, where small changes or potential errors can have cascading effects on the system, leading to unsafe operation conditions such as overloaded lines, or a complete breakdown of the system itself.

VIPER

Verifiable Reinforcement Learning via Policy Extraction (VIPER) [19] is a policy extraction method that improves upon DAgger to utilise the Q-function of the teacher model, and incorporate this into extracting interpretable decision trees. Similar to DAgger, a teacher-student architecture is used to obtain labels for the states with the respective actions that must be taken. Once a dataset and policy is initialised, the policy iteratively samples trajectories from its current state, and similar to DAgger, aggregates these datapoints into its existing dataset. After resampling from this dataset, the states are prioritised based on its importance to the expert policy, and a decision tree is trained on these state-action pairs, returning the best policy that imitates the expert as closely as possible.
VIPER is a relevant approach for working towards explainable reinforcement learning methods, as it provides a way to approximate the decision-making process through decision trees, which maintaining the performance as closely as possible to the expert policy. Nonetheless, there are its drawbacks; equal performance is not easily attainable, especially in policies where the teacher's neural network is extremely complex. Although proven to have equalled in performance in games like Atari Pong [19], this work was based on approximations and limitations in the tasks and actions possible. There still exists a research gap in understanding the complete potential of this approach in other complex domains, such as power grid operations.

S-REINFORCE

Although work in large neural networks has grown to predict actions in complex actions spaces with high accuracy, there has been a emphasis within the research community to develop neuro-symbolic policies [20, 21], where human-readable mathematical expressions determine how a policy learns, improving transparency into a model. One approach is S-REINFORCE [22], an intrinsically interpretable reinforcement learning algorithm which combines policy optimisation through neural networks with symbolic regression to learn an interpretable policy directly during training.
In this algorithm, symbolic regression and reinforcement learning are integrated into an iterative training procedure. Based on the standard REINFORCE algorithm [23], trajectories are generated within the environment, using discounted future rewards to update the policy for providing higher rewards. A neural network is used as the learning mechanism for mapping state to action probabilities. While the neural network trains on all the data across all trajectories collected, the symbolic component is introduced periodically, where the regressor aims to find a mathematical policy that approximates the neural network's policy. Genetic operators, such as crossover, mutation, and selection are used for this approximation mechanism [22]. The trained symbolic policy may also be used during training through importance sampling, correcting the policy-gradient estimate through the likelihood ratio. This ratio is a comparison of similarity between the neural network's policy and the symbolic regressor's policy. It should be noted that this is only useful if the likelihood ratio is high, since large differences between both policies can lead to unstable learning and

high variance.

2.2.2 Intrinsic Interpretability

While post-hoc methods develop an additional model that aims to explain pre-built models, they are not interlinked with the predictions themselves, leading to unreliable outputs or potential. To improve upon the transparency when developing glass-box models, intrinsic interpretability is favoured. For these models, the policy has interpretability built-in, and the model is able to explain its action prior to implementing it. There exists various different methods of making policies intrinsically interpretable. Previous work in this domain explored creating frameworks that generate symbolic policies to widen the scope of deep reinforcement learning [24, 25], converting policies into decision trees [26], using graphs and finite state machines [27], or logical expressions to develop both explainable and interpretable models. Recently, with the rise of large language models, their strengths have been adapted in various works to improve upon neuro-symbolic methods to provide textual explanations [28].
It should be highlighted that interpretability and explainability are not synonyms, albeit they do go hand in hand; the former refers to whether a learned policy is transparent and human-readable, and its decision-making process can be understood. On the other hand, the latter refers to whether the model has the capability to justify why a specific decision has been made, and what key inputs were critical in the action being chosen for the environment. The term to describe the minimal conditions and most significant inputs and contributors that are the foundation behind a certain output, action, or decision, are known as the prime implicants [29].

2.2.3 Interpretability Metrics

There are countless properties that are used to measure the value of interpretability and explainability within the field of explainable reinforcement learning. Molnar's [30] work on Interpretable Machine Learning delves deep into this. To summarise his work, the properties that can be used to measure explainable methods holistically are the model's expressive power, translucency, portability, and algorithmic complexity. These properties are also referenced in the works of Carvalho et al. [31] and Robnik et al. [32]. Expressive power refers to the architecture and structure of the model itself in order to convey its explanations. Translucency refers to how dependent the model is on utilising its inputs and parameters to be able to come up with an explanation. If policies are intrinsically interpretably, translucency is high. Portability refers to how versatile the method is, such that it can be applied to other domains. It should be noted that translucency and portability are inversely correlated; if an explainable model has high translucency, it has low portability, and vice-versa. Algorithmic complexity, as self-explanatory, takes into consideration the mathematical efficiency of the algorithm used to implement the explainable method. If looking particularly at individual explanations themselves, the properties that play a role and must be considered and/or measured are accuracy, fidelity, consistency, stability, comprehensibility, certainty, degree of importance, novelty, and representativeness [30]. Indubitably, we want to develop a glass-box model that is able to retain the strengths of black-box models, such as minimising false positives and false negatives, while ensuring that the glass-box model is closely aligned with the predictions of the black-box model - this is where accuracy and fidelity go hand in hand. On top of this, as with any reinforcement learning models, we require low variance and repeatability to ensure high reliability of the model, which is the reason why consistency and stability are highlighted. Comprehensibility refers to the human-readability aspect of the explanation, which is difficult to define or measure, as this is dependent on the context and domain in which the explainable model is being used. Representativeness the coverage of the explanation itself, and how much of the model it is able to explain. Certainty and novelty also go hand in hand, since it is extremely useful for a model to be able to provide its confidence metric on its own prediction of the most optimal action and its explanation. If during testing, a state or inputs unknown from testing come up (high novelty), the metric for certainty is sure to drop, and provides insights for users of the policy to know how useful the explanation and action might be. In domains like power grid operations, due to the complexity of the environments, this is extremely useful for power grid operators. Degree of importance refers to the number of prime implicants in a decision when predicting the next action.

2.3 Simulation Framework

For the purpose of this work, the RL2Grid environment [9] will be utilised as the benchmark reference for both assessment of feasibility as well as evaluation of the final contributions.

2.3.1 Grid2Op

Grid2Op [33] is an open-source Python-based framework that provides a realistic simulation framework for sequential decision-making in electrical power grid systems. It was developed as part of the ”Learning to run a power network” (L2RPN) challenge. It not only provides a modular tool that allows for the simulation of power grid operations, but facilitates the evaluation of various control strategies that enact realistic operating conditions within transmission grids. The fact that decisions are sequential means that at each timestep, the current state of the power grid is to be observed by an agent, which can take the necessary respective action.
A power network is created as part of an environment, which has multiple elements that can be modelled as part of it, which include generators, loads, power lines, substations, shunts, and storage units. Based on observations of the environment, which can be anything from the exact time to the active power flow at a specific power line, an agent can be instructed to take a respective action. These actions are undoubtedly constrained, given the nature of power grids; due to issues such as overloading and thermal limits, flow is limited heavily, and the agent must ensure that cascading failures do not occur due to overloading of buses. Therefore, it is vital

for the agent to not violate operational constraints. Additionally, depending on the type of event, failures can occur either in a deterministic and/or stochastic manner, which must be handled aptly by the agent.
Grid2Op provides a rich observation space containing information about line loadings, outputs from generators, load demands, configurations and connectivity of networks, amongst other characteristics. The modularity and flexibility that this framework provides made it the foundation for the RL2Grid benchmarking framework.

2.3.2 RL2Grid

RL2Grid [9] is an open-source, Python-based benchmarking environment designed specifically to standardise the evaluation and testing of various reinforcement learning algorithms within power grid operations. Designed based on the power simulation framework of Réseau de Transport d'Électricité France (RTE France), RL2Grid aims to provide reproducible experimental environments for comparing different approaches.
This benchmark consists of base grid environments of varying complexity, which is defined by the number of substations, lines, generators, and loads present in the system. It should be noted that all grids have a double bus system, where each component of the grid is connected to two other components. Table 2.1 below shows a subset of the available environments that we are taking into consideration for our work, which do not take include environments that are susceptible to stochastic events that cause disconnection within lines (opponents), nor does it include environments that have in-built batteries for supporting the power grid system. All these environments are purely for maintenance, which are deterministic disconnection of lines for set periods of time. Although the bus5 environment is not available through the original work, the author was kind enough to share this environment with us for our research and work for this thesis. These environments allow for the evaluation of scalability and robustness of reinforcement learning algorithms.

Table 2.1: Partial list of environments within RL2Grid and their properties. [9]

Environment	*No. of substations*	*No. of Lines*	*No. of Generators*	*No. of Loads*
bus5	5	8	2	3
bus14	14	20	6	11
bus36-M	36	59	22	37
bus118-M	118	186	62	99

Agents are typically evaluated on the survival rate metric, which measures the normalised proportion of timesteps that a grid remains operational [9]. A lower survival rate corresponds to a higher probability that the grid is likely to collapse, whereas a higher survival rate denotes long term stability in operation.
As part of the benchmark, implementations of several reinforcement learning algorithms have already been provided. The algorithms intrinsically provided are DQN,

PPO, Soft Actor-Critic (SAC), Lagrangian PPO (LAGR-PPO), and Twin Delayed Deep Deterministic Policy Gradients (TD3). As RL2Grid provides a controlled environment for testing of various algorithms and methods, as well as a solid foundation for investigating explainability and interpretability, it is well suited for this work on researching explainable reinforcement learning.

Topology

Within RL2Grid, there exist two different types of action spaces, namely topology and redispatch [9]. While redispatch refers to continuous actions, a topology action space consists of discrete actions, where agents can take decisions on lines or buses-component connections. Due to the combinatorial nature of the action spaces due to the number of potential switching configurations, this grows exponentially as the grid size increases. This raises the challenge for developing strong interpretable policies that are able to scale, making it suitable for this line of work. Although algorithms like PPO are able to learn effective policies with a high survival rate, there still exists the issue of scalability and a lack of interpretability and explainability, which serves as the knowledge gap that we aim to take advantage of and pursue.

2.4 Interpretability Evaluation

Undoubtedly, most work requires a human-in-the-loop approach to not just evaluate frameworks and methods against one another, but to also benchmark against subject matter experts and operators who have substantial experience in their domain.
As per the work of Doshi-Velez and Kim [34], their proposed taxonomy to evaluate interpretable model can be classified under 3 distinct methods; application-based, human-based, and function-based. An application-based evaluation includes a human expert working within a real environment for comparison against the interpretable model. A human-based approach for evaluation works best when the environment is challenging to recreate or replicate, whether this is due to the complexity of the environment, financial constraints, or other issues. Here, solely the quality of the explanation itself is evaluated instead of the agent's behaviour in the entire environment. The least expensive and more generic option is the function-based evaluation, where the model has already been evaluated and validated by a human. In such cases, 'proxies' are used for evaluation purposes. For example, this could be the depth of a decision tree.

Chapter 3

Preliminary Results

For the initial aim of the project of understanding what is possible within power grid operations using explainable methods, techniques such as DAgger, VIPER, and S-REINFORCE were implemented within the RL2Grid framework to provide a useful starting point for understanding whether black-box RL controllers can be successfully converted into more interpretable policies. Since the emphasis is on the generation of decision trees, this was done subsequently after ensuring that a high survival rate was feasible after implementation.

Table 3.1 below shows the preliminary results of the three methods on environments of three different sizes. Across these tests, it was observed that a combination of the VIPER method trained on a DQN oracle showed great promise, showcasing the best survival rate. However, as initially hypothesised, due to the complex nature of the environments, and an emphasis of keeping the depth of the tree as minimal as possible, scalability stood out as an issue for larger environments. Due to the large combinatorial action space within a topology network, the depth of the tree increases, reducing the interpretability of the model. Since a shallow tree can only capture simple rules, there exists a challenge in understanding not only how neural policies can be distilled into decision trees, but also how interpretable controllers can be made scalable without losing information.

On the other hand, S-REINFORCE showcased the weakest performance across the

Environment	Method	Oracle	Best Survival Rate
Bus5	VIPER	DQN	100.000%
Bus5	VIPER	PPO	100.000%
Bus5	DAgger	PPO	100.000%
Bus5	S-REINFORCE	–	47.073%
Bus14	VIPER	DQN	87.855%
Bus14	VIPER	PPO	63.579%
Bus14	DAgger	PPO	54.177%
Bus14	S-REINFORCE	–	5.659%
Bus36-M	VIPER	DQN	53.399%
Bus36-M	VIPER	PPO	14.351%
Bus36-M	DAgger	PPO	4.875%
Bus36-M	S-REINFORCE	–	8.249%

Table 3.1: Best survival rates achieved across different environments and three algorithms.

three methods, suggesting that further finetuning of hyperparameter configuration and setup may be required for improvement within this domain. There also exists an anomaly in the S-REINFORCE results only, where the best survival rate for bus14 is lower than that of bus36-M. This could be due to the complexity of the action space, and the symbolic regressor unable to mathematically approximate the neural network's policy.

Although not as weak as S-REINFORCE, since VIPER with a PPO oracle and DAgger with a PPO oracle achieved lower survival rates compared to VIPER with a DQN oracle, this clearly indicates that the choice of oracle has a significant effect on the quality of the extracted policy. This was further reaffirmed during testing, where incompletely trained expert oracles would lead to drastically poor survival rates, which would have a cascading impact when training the interpretable policies. Additionally the quality of the environment generated by the random seed chosen when training both the expert oracle as well as the interpretable method had an impact on whether a high survival rate was reached or not. In higher environments, this inconsistency was further exacerbated.

Since scalability was seen as a major factor, subsequent testing focused solely on the bus5 environment itself, since these trained policies were able to reach a 100% survival rate. Decision trees were generated through policy distillation and plotted. The following images, although do not show the complete tree, showcase a small portion to highlight the information gathered at each node.

As shown in the figures above, the nodes showcase the information pertaining to its respective observation (in other words, the decision condition), the Gini impurity at that node, the number of samples, the distribution of classes (actions), and the action selected by the tree. For example, the top node in the snapshot of Figure 3.1 states observation_24 and class_5, which refers to the active power flow at powerline 4's origin in megawatts, and the changing of busbars at powerline 4's origin, respectively.

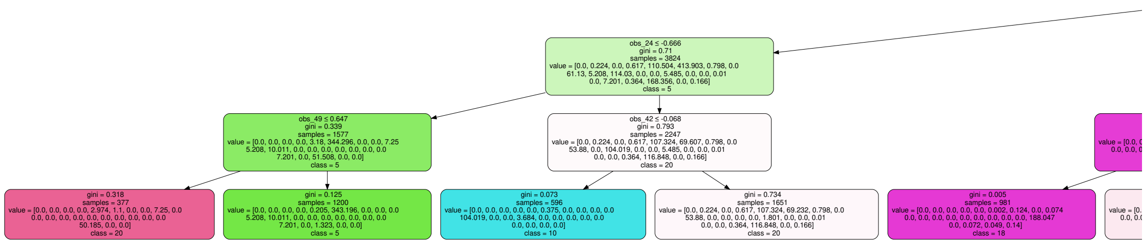

Figure 3.1: Snapshot of the full VIPER decision tree for Bus5.

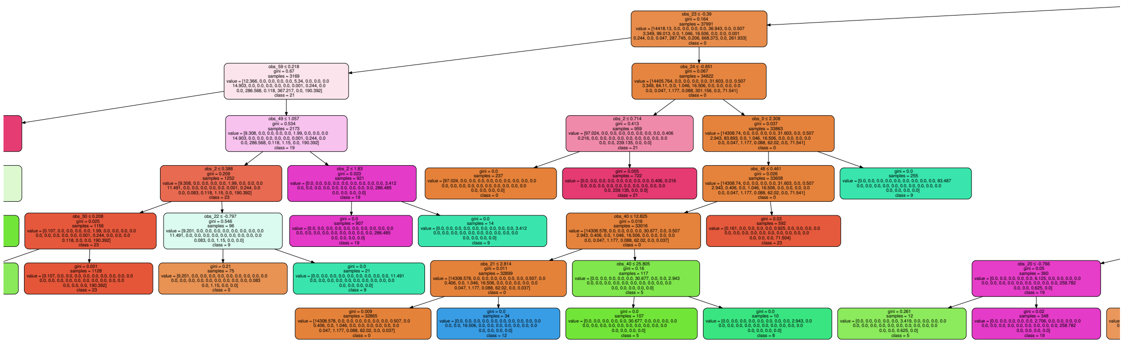

Figure 3.2: Snapshot of the depth-8 pruned VIPER tree for Bus5.

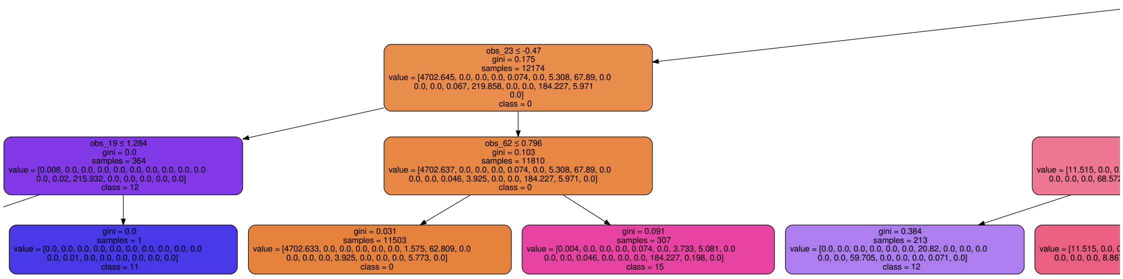

Figure 3.3: Snapshot of the depth-4 distilled VIPER tree for Bus5.

Since the depth and width of a decision tree plays a crucial role in its interpretability, we aim to keep this as minimal as possible. Various techniques, such as pruning of higher-depth trees, or the distillation into singular, simple trees were experimented with, and showcased similar results in retaining the perfect survival rate within the environment. Figure 3.2 showcases a portion of a pruned tree of depth 8. Figure 3.3 showcases a portion of a decision tree that was distilled successfully into a depth of 4, while maintaining a 100% survival rate. This shows feasibility of decision trees on small environments, but requires further work in ensuring scalability.
A key observation from this preliminary work is the potential disagreement between offline fidelity and online operational performance. We suspect that although a decision tree policy extracted from a neural controller is able to imitate an oracle policy on a collected network, it might still achieve a lower survival rate when deployed within the environment itself. Thus, this distinction raises a key point that although

survival rate serves as a great metric for understanding whether a policy can maintain grid operation, it should not be used as the only metric for validating that an interpretable controller is safe and reliable.
Therefore, as will be detailed in the next chapter, the subsequent work requires further analysis of the decision tree structure to determine whether it remains interpretable within large environments.

Chapter 4

Project Plan

The remainder of this project will be focused on developing a scalable method for training and extracting interpretable controllers for power grid topology-based environments. Therefore, the main focus for the next stage is to investigate whether decision tree-based controllers can preserve their interpretability while achieving strong survival rate performance in complex environments.
The main research question for the remainder of this project is:

> Can a decision tree policy distilled from a trained reinforcement learning oracle be further optimised to improve survival rate in power grid topology control, while preserving the interpretability benefits of tree-based controllers?

This research direction has been motivated by a gap between two existing methods of interpretable reinforcement learning. Methods like Decision Tree Policy Optimisation (DTPO) [35] are great at optimising decision tree policies, but have not been tested on scalability due to large batch requirements and a lack of a strong teacher policy for initialisation. On the other hand, policy distillation methods are benefitted by the presence of a teacher policy for extracting decision trees, yet they are significantly limited by the quality of the teacher policy, and do not provide an opportunity for further improvements. A distillation technique is the use of asymmetric differentiable decision trees, which allow for improved efficiency and interpretability, and consider scalability (as only useful select branches are expanded), but if the teacher model is weak, this creates an equally weak student model. This showcases an important research gap between the lack of amalgamation of distillation and optimisation.
Thus, the proposed idea is to implement and test a hybrid approach, in which a decision tree will be first distilled from a strong reinforcement learning oracle. This decision tree will then be fine-tuned using policy optimisation directly within the power grid environment generated by RL2Grid. The aim is to combine the benefits of learning an initially strong policy from an equally strong oracle with the performance improvements that will be obtained through policy optimisation. This is necessary for scaling, as RL2Grid showcases that even the best reinforcement learning algorithms like PPO do not exceed a survival rate of 29% in bus environments

greater than or equal to bus36-M [9]. Thus, this hybrid approach aims to combine the best of both worlds to achieve both performance and interpretability.

For the planned methodology, a strong reinforcement learning oracle will first be trained or selected for each environment. This oracle may be based on an existing DQN or PPO policy, depending on which achieves stronger survival performance based on the environment chosen. Then, trajectories will be collected from the oracle in the target Grid2Op environment. After this, an interpretable decision tree policy will be distilled from the oracle using the collected state-action pairs. Based on the results of the preliminary results, there will be an emphasis to ensure the tree is not forced to be symmetrical. This distilled policy will be evaluated as a standalone controller, checking whether the tree can imitate the oracle well and whether readability is kept. The metrics that will be evaluated at this stage include survival rate, fidelity to the oracle, tree depth, number of leaves, and the distribution of selected actions.

For the next step, this distilled tree will be used as an initialisation for a policy optimisation stage, where the tree will be allowed to directly interact with the RL2Grid environment. This stage will focus on investigating whether the policy of the tree can be improved further. This resulting policy will be compared against the original oracle as well as the initially distilled tree, as well as existing baselines reinforcement learning algorithms present within RL2Grid. The goal of this step will be to observe and understand whether optimisation can recover performance that may have been lost through distillation, while keeping interpretability of decision trees. A successful outcome will be a decision tree generated that is able to show meaningful improvement in performance over the purely distilled tree, maintaining a high survival rate, while keeping human-readability. The most optimal outcome would be that this decision tree policy is able to match or exceed the performance of the oracle itself. This will be extremely important in highly complex environments; as aforementioned, existing baseline algorithms already struggle to reach acceptable survival rates.

This proposed hybrid plan is novel; to the best of our knowledge, this approach has not been implemented before, let alone within the context of power grids. It is also understandably ambitious, given the time constraints of the project, and the risks associated with this hybrid approach with regards to unstable policy training of the tree after distillation, the computational cost of running the decision tree within the environment, and ensuring that the trees do not grow significantly in size whereby interpretability is lost. If these issues occur and persist, as a contingency plan, the goal of the project will pivot towards using a post-hoc concept-based explanation framework for the distilled tree policies.

This work will shift towards understanding the prime implicants and concepts that explain the actions taken in states within the environment. This is inspired by the work of Patel et al. [36], where an explainability framework, Agua, is introduced to "[explain] a model's decision using high-level, human understandable concepts." [36] Agua uses two different types of mapping, namely concepts and outputs, that allows a surrogate model to learn from an existing black box model to map the internal features onto a concept space, and then mapping this space onto prediction

actions taken in the environment. This allows for a higher-level understanding of why actions are taken, which is human-readable and interpretable.
Since this framework and similar post-hoc methodologies have not been implemented previously within power grid operations, it serves as a research gap that we aim to investigate further. In this work, our existing results on generating decision trees and extracting meaningful grid features may be used for concept-based representation, such that the policy for generating a decision tree can be used to either improve upon or steer the learning of the trees in an iterative manner. We aim to implement and train this surrogate model of Agua within the RL2Grid environment to provide explainability for the actions taken within a power grid system.
Within the month of June, the first two weeks will be catered towards the individual code implementation of the asymmetric distilled decision tree, as well as the policy optimisation method for training decision trees, after which there will be an emphasis on combining these two code bases for implementing the aforementioned hybrid approach. The second half of June will then be focused upon testing, checking performance improvements (if any) as well as carefully evaluating the interpretability of the new approach. Initial tests will check feasibility and functionality within the bus5 environment, before observing scalability within the bus14 and bus36-M environments. If this hybrid approach shows promising results, the month of July will be used for hyperparameter optimisation and further improvements to the codebase, as well as the formalisation of the hybrid approach.
If decision trees are observed to not stay interpretable, or the survival rates observed are not close to that of the purely distilled decision trees or the teacher models, the focus of the project will shift towards the contingency plan of pursuing the concept-based post-hoc method. This decision on the main direction of the project will be made at the end of June, depending upon the viability of the hybrid asymmetric decision tree optimisation approach. As the Agua [36] framework has been proven in different domains, it shows promise it can also be implemented within the power grid domain.
During the month of July, the first week will be spent on implementing the code for the Agua framework, adapting it for use within the RL2Grid environment. The second week will be used for testing, to understand the mapping and concepts generated, after which state-of-the-art LLMs will be used for generating the explanations for these concepts. The last two weeks of July will be used for testing purposes, obtaining data and confirming validity of the framework in environments of differing sizes, which at least will include bus5, bus14, bus36-M, and bus118-M. Regardless of the approach that will be finalised by the beginning of July, the month of August will be spent consolidating the experimental results, conducting final evaluations and ablation studies. The final figures and tables required for the final submission will be generated in the first two weeks of August. The last two weeks will be dedicated solely to writing, editing, and finalising the thesis submission.

Chapter 5

Evaluation Plan

In this project, RL2Grid serves as the primary experimental framework for the evaluation for interpretable reinforcement learning methods developed and tested. Since this framework uses survival rate as its primary metric, this will serve as the first evaluation and operational metric, as the central objective is to prevent grid collapse and maintain stable operation over time.

The second evaluation metric is fidelity to the oracle policy. Fidelity will be measured to ensure that the interpretable student policy selects the same actions as the teacher policy on a dataset of observed states. If some states are more important than others, especially in higher complexity environments, weighted fidelity will be used. The formula for this will be as follows:

$$\text{Weighted Fidelity} = \frac{\sum_{i=1}^{N} w(s_i)\mathbb{I}\left[\pi_{\text{tree}}(s_i) = \pi_{\text{oracle}}(s_i)\right]}{\sum_{i=1}^{N} w(s_i)} \tag{5.1}$$

Interpretability will undeniably be part of the evaluation criteria. For decision trees, this will be measured through tree depth, number of leaves, number of decision nodes, and number of features used. The focus will be to minimise these metrics as best as possible, to improve human-readability. Scalability will also be tested, by comparing the final method across environments of different sizes, as well as evaluation computational cost. If successful, stochastic events and the use of batteries will also be incorporated into the environments to show feasibility regardless of constraints.

Since we aim to develop a hybrid method, this will certainly be evaluated against several baselines, which include the original reinforcement learning oracle, as well as purely distilled decision tree policies that were generated and included as part of the preliminary results. As there are multiple components to the hybrid approach we propose in this work, an ablation study will be conducted to understand the contribution of each specific component, to help understand where exactly the improvement within the model comes from.

Each method will be evaluated over multiple episodes. Due to the CPU-bound nature of RL2Grid, consistency checks across multiple random seeds will only be conducted where computationally feasible. As the initial grid condition and actions taken can have a trickle-down effect on the entire environment, results will therefore be re-

ported using average best survival rate, and where possible, variance or standard deviation across evaluation runs.
The main focus for evaluation will be to observe the trade-off between performance of a model and the interpretability present. The goal is to maintain reasonable survival performance while ensuring that the tree stays compact enough for the presence of human-readable characteristics.

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